Allen R. Majewski, PhD

I take production LLM systems from ambiguous requirements to safe rollout, build the apparatus the team operates them through, and resolve systemic accumulation nobody else has patience to diagnose.

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Experience

Deloitte Chicago, IL (Remote)
Senior AI Systems Engineer February 2023 — Present

Sole maintainer and release gatekeeper of the VA records-processing pipeline (Django / Celery / Redis / S3). Supports three engineers; owns CI/CD, regression testing, and shadow deployment. Scaled production throughput from ∼100k to ∼3M pages/month over 6 months; on track to ∼7.5M pages/month within 3 more months (∼75× ramp).

Production ML and LLM systems

- Raised redaction recall from ∼20% to ∼96% across a multi-stage PII pipeline. Residual misses concentrated in OCR-degraded edge cases; page-level risk model reaches ∼90% recall, enabling ∼95% reduction in manual review via sparse (∼1%) page-level flagging and ∼10% request-level autopass.
- Architected a hybrid redaction stack combining label-guided, LLM-based, and validated regex methods with gated stamp detection, reaching production reliability neither approach alone achieved. LLM inference via Amazon Bedrock (on-demand and batch).
- Designed and shipped a vision-LLM template extractor replacing OCR + coordinate-box + search-word matching for structured VA forms. Claude Sonnet 4.6 vision, feature-flagged, shadow-compared against legacy; per-form rollout with explicit cutover criteria.
- Shipped all high-consequence logic via shadow deployment, controlled experiments, and progressive rollout. Built the release pipeline from scratch.

Load-bearing artifacts authored

- **inspector —** ∼7000-line Python CLI framework for operating the redaction pipeline. Typed Request / Document / Page hierarchy, django-style `.filter()` on typed RedactionSet, diff / compare / health subcommands, hand-authored groff man pages. First-class machine interface used in production ops and by AI agents.
- **Guards manifesto —** formal taxonomy of the redaction pipeline (substrate → producer → guard → oracle; invariant pipeline gating → produce → guard → score → pick → emit). Adopted as team convention; checked in as canonical repo reference.
- **Memory system for AI-assisted development —** persistent, typed memory (user / feedback / project / reference) governing how coding agents operate on this codebase. Five-criteria policy (true, maps to external knowledge, load-bearing, stable, non-derivable from repo) with explicit negative criterion.

Named refactor arcs (cross-cutting cleanups shipped under formal release gating)

- **MonsterMoige —** removed a four-way code fork across two repositories. Swung imports in 82 production files, deleted ∼3000 lines of duplicated code, stabilized an 832-test suite by eliminating module-level `os.environ` side effects that had silently supplied dependencies to unrelated tests.
- **LGRefactorooney / FNRRefactorooney —** extracted ∼1650-line and ∼920-line god-class redactors into domain packages (redactor / guards / score / gating / prompts / patterns / utils) with module-level guard functions, preserving semantics under full regression.
- **StampPackagify / RekognitionLift —** packaged the last flat-file redactor into the canonical layout; lifted the AWS Rekognition client to a shared parent-level module; promoted the guards manifesto to v1.1.

AI-assisted engineering practice

- Orchestrate multiple coding agents (Claude Code, Copilot) at the level of design intent — authoring manifestos, naming conventions, and memory policies the agents operate against. Built the memory directory and convention files explicitly as infrastructure for this.

FlickPlay Los Angeles, CA (Remote)
Machine Learning Engineer (Contract) November 2022 — February 2023

- Built analytics platform from raw application DB to a feature space for user-behavior modeling. Derived first-order engagement and retention signals; reported directly to CEO.

KPMG US Chicago, IL
Senior Associate, Data Scientist (Contract) April 2022 — November 2022

- Shipped two storage-optimization solutions to Azure production via linear-programming models (PuLP) and industrial 3D bin-packing middleware for smart-warehouse robotic picking.

UncommonX Chicago, IL (Remote)
Lead Data Scientist August 2020 — February 2022

- Built the data-science function from zero. Invented the UncommonX cyber risk score. Shipped representation-learning pipelines (IP2Vec), anomaly detection, autoencoders, and temporal models over Kafka / NetFlow streams as production REST APIs consumed by engineering teams.

Rhombus Power, Inc. Moffett Federal Airfield, CA (NASA Research Park)
Research Engineer / Data Architect February 2019 — February 2020

- End-to-end delivery across architect / engineer / scientist roles. Real-time pipelines curating >4 TB of unstructured data on AWS RDS / EC2; parallel compute (multiprocessing, Spark, MPI, SLURM) for government clients.

University of Florida Gainesville, FL
Graduate Research Assistant, Computational Physics (Cheng Group) August 2015 — May 2019

- PhD in computational materials physics. Authored an OO Python API for Quantum ESPRESSO (job creation, parsing, validation, DB bindings) over high-throughput massively-parallel HPC simulations.
- First-author preprint on first-principles DFT methodology for temperature-dependent NQR parameters in molecular crystals (arXiv:1903.10097, with Cheng / Sullivan groups). Combines MD with DFT to predict thermal behavior of quadrupolar coupling without model-dependent assumptions about intermolecular potentials.

Selected Writing

- **The Cost of a 401k Loan, Measured in Time** — a phase-transition treatment of personal finance: a single ODE, $dP/dt = gP + c$, two engines (labor and capital), and a closed-form identity-preservation formula whose denominator vanishes at the labor-capital crossover. Frames retirement-system design as a robustness-vs-efficiency trade-off.

Education

University of Florida 2019
PhD, Physics — Dissertation: Density Functional Theory Calculations of Temperature-Dependent NQR Parameters in Molecular Crystals

University of Florida 2015
MS, Physics

University of Florida 2010
BS, Physics | *BA, Mathematics*

Technical

- **Languages:** Python 3, SQL, Bash.
- **Cloud:** AWS (Bedrock, SageMaker, ECS, EC2, RDS, S3), Azure, Azure DevOps.
- **Infra:** Docker, Kubernetes, Argo CD.
- **Backend:** Django, Celery, Redis, MySQL, MongoDB.
- **ML:** scikit-learn, NumPy, SciPy, pandas, LangChain, LangGraph.
- **Data:** Kafka, Elastic Stack, Airflow, MPI, Apache Spark, SLURM, PuLP.